



# Information on Rapid Molecular Assays, RT-PCR, and other Molecular Assays for Diagnosis of Influenza Virus Infection



For Health Care Providers

APR. 30, 2026

## PURPOSE

- Guidance for Clinicians on the Use of RT-PCR and Other Molecular Assays for Diagnosis of Seasonal Influenza Virus Infection.

## Background

Molecular assays available for detecting seasonal influenza A and B virus infection include Reverse Transcription-Polymerase Chain Reaction (RT-PCR), and other nucleic acid amplification tests. These tests can detect influenza viral RNA or nucleic acids in respiratory specimens with high sensitivity and high specificity. Notably, the detection of influenza viral RNA or nucleic acids by molecular assays does not necessarily indicate detection of viable virus or on-going influenza viral replication. Available FDA-cleared molecular assays are listed in the **Table FDA-cleared or FDA-authorized RT-PCR Assays and Other Molecular Assays for Influenza Viruses**. A list of [FDA-cleared and FDA-authorized influenza diagnostic tests, including nucleic acid amplification assays](#), for use in clinical settings or for home use is available.

Rapid molecular assays are molecular influenza diagnostic tests that can detect seasonal influenza viral RNA or nucleic acids in upper respiratory tract specimens in approximately 15-30 minutes. One platform uses isothermal nucleic acid amplification and has high sensitivity, and yields results in 15 minutes or less. Other platforms use RT-PCR and produce results in approximately 20-30 minutes. Reported sensitivities of available rapid molecular assays range from 66-100%. As with other molecular diagnostic tests, if antiviral treatment is clinically indicated (e.g., patients at increased risk of complications, outpatients with progressive or severe disease, hospitalized patients) initiation of empiric antiviral treatment should NOT be withheld from patients with suspected influenza while awaiting testing results during periods of peak influenza activity in the community when the likelihood of influenza is high. More information about antiviral treatment of influenza is available at [Antiviral Drugs, Information for Health Care Professionals](#).

## KEEP READING

[Influenza Antiviral Medications: Summary for Clinicians](#)

Some multiplex molecular assays are able to detect and discriminate between infections with influenza A and B viruses; other tests can also identify specific seasonal influenza A virus subtypes, for example A(H1N1)pdm09, or A(H3N2). FDA-cleared rapid molecular assays can provide results in 15-30 minutes, and some are CLIA-waived for point-of-care use. Other molecular assays can provide results in 45-80 minutes or several hours, depending upon the assay. Some FDA-cleared and FDA-authorized multi-pathogen molecular assays are available for use in clinical settings or for home use that can detect seasonal influenza A and B viruses and other respiratory pathogens (e.g., SARS-CoV-2, RSV).

## How to Use

### Clinical Decision Making

- Influenza testing is not needed for all outpatients with signs and symptoms of influenza to make antiviral treatment decisions (See [Figure 1, Figure 2](#)). Once influenza activity has been identified in the community or geographic area, a clinical diagnosis of influenza can be made for outpatients with signs and symptoms consistent with suspected influenza, especially during periods of high influenza activity in the community. Rapid molecular assays that can produce results in approximately 15-30 minutes, and other molecular assays that can detect influenza viral RNA or nucleic acids in approximately 45-80 minutes are available in hospitals (see **Table FDA-cleared RT-PCR Assays and Other Molecular Assays for Influenza Viruses**). The Infectious Diseases Society of America (IDSA) recommends use of rapid influenza molecular assays over rapid influenza diagnostics tests (RIDTs, for antigen detection) for influenza testing of respiratory specimens of

outpatients. Consult the [IDSA Influenza Clinical Practice Guidelines](#) for recommendations on influenza testing and information on interpretation of testing results

- Clinicians should be aware of the approved clinical specimens for the molecular influenza assay being used.
- Molecular influenza testing is recommended for all hospitalized patients with suspected influenza.
- If treatment is clinically indicated, antiviral treatment should NOT be withheld from outpatients or hospitalized patients with suspected influenza while awaiting influenza testing results.
- Since results from molecular influenza assays may not always be available when initial therapy decisions must be made, influenza antiviral treatment should be started as soon as possible because the greatest clinical benefit is when treatment is initiated as close to illness onset as possible, especially for hospitalized patients and outpatients at high risk of serious complications.

#### KEEP READING

[Influenza Antiviral Medications: Summary for Clinicians](#)

## Influenza Testing of Hospitalized Patients

- Hospitalized patients with suspected influenza without lower respiratory tract disease should have upper respiratory tract specimens collected for influenza testing by a molecular influenza assay. The Infectious Diseases Society of America (IDSA) recommends use of RT-PCR or other molecular assays for detection of influenza viruses in respiratory specimens of hospitalized patients. Consult [IDSA Influenza Clinical Practice Guidelines](#) for recommendations on influenza testing and information on interpretation of testing results.
- More information about antiviral treatment of influenza is available at [Influenza Antiviral Medications: Summary for Clinicians](#).
- Collection of lower respiratory tract specimens from hospitalized patients with suspected influenza and pneumonia can be considered for influenza testing by RT-PCR and other molecular assays if influenza testing of upper respiratory tract specimens is negative and if positive testing would result in a change in clinical management. Hospitalized patients with suspected influenza and respiratory failure on mechanical ventilation can have an endotracheal aspirate specimen collected for influenza testing by RT-PCR if a laboratory diagnosis of influenza has not been determined. Bronchoalveolar lavage fluid, if collected for other diagnostic purposes, can also be tested by RT-PCR or other molecular assays for influenza viruses. Currently, only the CDC RT-PCR assay is FDA-cleared for lower respiratory tract specimens; this test is available only at qualified public health laboratories (see **Table FDA-cleared RT-PCR Assays and Other Molecular Assays for Influenza Viruses**). Clinicians may elect to order other FDA-cleared assays for off-label use in evaluating lower respiratory tract specimens. Performance of these assays for these specimens has not been evaluated by FDA; however, these assays may be more readily accessible at some institutions.

#### KEEP READING

[Influenza Antiviral Medications: Summary for Clinicians](#)

## Detecting Institutional Influenza Outbreaks

- Molecular assays such as RT-PCR are particularly useful to identify influenza virus infection as a cause of respiratory outbreaks in institutions (e.g., nursing homes, chronic care facilities, and hospitals).
- Positive results from one or more ill persons with suspected influenza can support decisions to promptly implement prevention and control measures for influenza outbreaks. Clinicians should be aware of requirements from their public health authorities regarding prompt notification of any suspected or confirmed institutional influenza outbreaks, and when respiratory specimens should be collected from ill persons and sent to a public health laboratory for laboratory confirmation of influenza.
- [Viral Respiratory Pathogens Toolkit for Nursing Homes](#) is available.
- [Recommendations for control of institutional influenza outbreaks by the Infectious Diseases Society of America](#) is available.

## Detecting Novel Influenza A Cases

- Molecular assays like RT-PCR identify influenza A and B viruses by targeting conserved gene regions. Some assays detect influenza A or B but do not specify subtypes, therefore it's not possible to identify novel influenza A viruses from these tests. The novel influenza A virus novel strains are antigenically and genetically different from current seasonal strains, often originating from zoonotic transmission from avian or swine species to humans.
- Some FDA-cleared devices detect influenza A or B viruses and can identify specific influenza A hemagglutinin genes, determining subtypes like A(H1N1)pdm09 or A(H3N2). These assays identify currently circulating strains and may also detect influenza A viruses without specified subtypes, known as "unsubtypeables." Such unsubtypeables could indicate novel influenza A virus infections, potentially representing new or unusual strains.

- Clinicians and laboratorians using molecular assays that are capable of detecting all currently circulating seasonal influenza A virus subtypes, for example A(H1N1)pdm09, or A(H3N2), and who identify an "unsubtypeable" result (i.e., influenza A with no subtype detected), should contact their state or local public health laboratory immediately for additional testing to determine if the infection is due to a novel influenza A virus.
- Clinicians and laboratorians using molecular assays that detect influenza A viruses that do not specifically identify currently circulating seasonal influenza A virus subtypes, for example A(H1N1)pdm09, or A(H3N2), and who suspect novel influenza A virus infection should contact their state or local public health laboratory immediately for additional testing to determine if the infection is due to a novel influenza A virus.

## Factors Influencing Results of Molecular Assays

Many factors can influence influenza testing results. Influenza viral shedding in the upper respiratory tract generally declines substantially after 4 days in immunocompetent patients with uncomplicated influenza, although infants and young children may have detectable influenza viruses for longer periods. Patients with lower respiratory tract disease may have prolonged influenza viral replication in the lower respiratory tract. Immunosuppressed patients and persons receiving systemic corticosteroids who have lower respiratory tract disease can also have prolonged influenza viral replication in the lower respiratory tract. Molecular assays can detect influenza viral RNA (positive results) for a longer duration than other influenza tests (e.g., antigen detection – immunofluorescence or rapid influenza diagnostic tests). Although molecular assays have high sensitivity, negative results can occur in patients with influenza for multiple reasons, so negative molecular assay RT-PCR results may not always exclude a diagnosis of influenza. If clinical suspicion of influenza is high, antiviral treatment should continue in patients with severe illness or [at increased risk for complications](#) while additional respiratory specimens are collected, and further influenza testing is performed. Factors that can influence influenza testing results are:

- Time from illness onset to collection of respiratory specimens for testing
- Respiratory specimens should ideally be collected as early as possible (ideally less than 4 days after illness onset when influenza viral shedding is highest) in persons without lower respiratory tract disease and tested as soon as possible. Molecular assays may be able to detect influenza viral RNA or nucleic acids in respiratory tract specimens longer than other tests can detect influenza viruses (e.g., after 72 hours from illness onset).
- Source of respiratory specimens tested and specimen handling.
- The best upper respiratory tract specimens to detect influenza viral RNA by RT-PCR and other molecular assays are nasopharyngeal swabs, washes or aspirates; other acceptable specimens are a nasal and/or throat swab. A swab with a wood shaft should not be used for respiratory specimen collection because it may interfere with RT-PCR and other molecular assays. Clinicians should be aware of the approved clinical specimens for the molecular assay being used and what type of swabs are recommended for use with the assay as included in the manufacturer's instructions included in the assay.
- Hospitalized patients with lower respiratory tract disease may have prolonged lower respiratory tract influenza viral replication compared to the upper respiratory tract. In patients with lower respiratory tract disease, lower respiratory tract specimens should be collected and tested if influenza is clinically suspected and testing of upper respiratory tract specimens is negative. For critically ill patients with suspected influenza, even when testing by RT-PCR or other molecular assays is negative, consideration should be given to collecting additional respiratory specimens from multiple sites, especially lower respiratory tract (endotracheal aspirate, or bronchoalveolar lavage – if clinically indicated for other diagnostic purposes) and re-tested for influenza viruses by RT-PCR or other molecular assays. Antiviral treatment should be continued in such patients pending additional influenza testing.
- Time from specimen collection to testing.
- If testing is delayed or is done at a facility other than where the patient is hospitalized, specimens should be placed in sterile viral transport media, consistent with test specifications, and refrigerated until transported to the laboratory for testing as soon as possible. Freezing and thawing should be avoided or minimized to avoid degradation of influenza viruses if viral culture will be performed.
- Adherence to manufacturer's instructions, including acceptable specimens, handling, and storage and processing, should be followed to achieve optimum test performance.
- Manufacturer's instructions, including acceptable specimens, handling, and storage and processing, should be followed to achieve optimum test performance. Deviations from recommended procedures may result in false negative results.

Collect respiratory specimens ideally within 4 days of illness onset for optimal influenza detection, especially in patients without lower respiratory tract disease. Test as soon as possible. Molecular assays can detect viral RNA longer than other tests, even after 72 hours.

Hospitalized patients with lower respiratory tract disease may have prolonged influenza replication in the lower tract. Test lower respiratory specimens if influenza is suspected and upper tests are negative. For critically ill patients, even with negative RT-PCR results, consider additional specimens from multiple sites, particularly the lower respiratory tract, and re-test for influenza. Continue antiviral treatment while awaiting further results

Hospitalized patients with lower respiratory tract disease may have prolonged influenza replication in the lower tract. Test lower respiratory tract specimens if influenza is suspected and testing of upper respiratory tract specimens are negative. For critically ill patients, even with

negative RT-PCR results, consider additional specimens from multiple sites, particularly the lower respiratory tract, and re-test for influenza. Continue antiviral treatment while awaiting further results.

## Interpreting test results

Sensitivities and specificities of RT-PCR and other molecular assays that have been cleared by the FDA for diagnostic use are very high compared to other FDA-cleared assays that use different methods. However, even with RT-PCR, false negative results can occur due to improper or poor clinical specimen collection or from poor handling of a specimen after collection and before testing. A negative result can also occur by testing a specimen that was collected when the patient is no longer shedding detectable influenza virus. False positive results, although rare, can occur (e.g., due to lab contamination or other factors). Information on interpretation of influenza testing results is also available in the [IDSA Influenza Clinical Practice Guidelines](#).

- Negative result
  - A negative result means that there is no evidence of influenza viral RNA or nucleic acids in the respiratory specimen tested. For hospitalized patients, especially for patients with lower respiratory tract disease, if no other etiology is identified and influenza is still clinically suspected, additional specimens should be collected and tested, and antiviral treatment should be initiated or continued.
- Positive result
  - A positive result indicates detection of influenza viral RNA or nucleic acids in the respiratory specimen tested, confirming influenza virus infection, but does not necessarily mean infectious virus is present or that the patient is contagious.
  - A positive result on testing of an upper respiratory tract specimen in a person who recently received intranasal administration of live attenuated influenza virus vaccine (LAIV) may indicate detection of vaccine virus. LAIV contains influenza virus strains that undergo viral replication in upper respiratory tissues of lower temperature (e.g., nasal passages) than internal body temperature. Since the nasal passages are infected with live influenza virus vaccine strains during LAIV administration, sampling the nasal passages within a few days after LAIV vaccination can yield positive influenza testing results. It may be possible to detect LAIV vaccine strains up to 7 days after vaccination, and in rare situations, for longer periods.
  - Influenza molecular assay interpretation will depend on the individual test that is performed. For example, a negative result from an influenza molecular assay that only detects influenza A virus and the A(H1N1)pdm09 subtype does not preclude infection with influenza B virus. Clinicians can consult the package insert for detailed description of each FDA-cleared test and what the result may or may not signify.

## Advantages/Disadvantages of Molecular Assays

### Advantages:

- Rapid molecular assays and some commercially available molecular assays can produce results in a reasonable time period to inform clinical management (ranging from approximately 15-30 minutes to less than 1.5 hours).
- Molecular influenza assays are more sensitive and specific for detecting influenza viruses than other influenza tests (e.g., antigen detection tests such as rapid influenza diagnostic tests or immunofluorescence assays, and viral culture).
- Because both sensitivity and specificity of molecular influenza assays to detect influenza viruses are high, the likelihood of a false positive or false negative result is low and therefore, the interpretation of the result is less impacted by the level of influenza activity in the community.

### Disadvantages:

- Results of some RT-PCR and other molecular assays may not be available in a clinically relevant time frame to inform clinical management decisions.
- RT-PCR and other molecular influenza assays may not be available in all outpatient or emergency room settings. For hospitalized patients, these assays are not always available on-site.
- Respiratory specimens may need to be sent to a public health laboratory or commercial laboratory for RT-PCR. Therefore, although RT-PCR assays can yield results in 4-8 hours, the actual time to receive results may be substantially longer.
- Most FDA-cleared molecular influenza assays are not approved to test lower respiratory tract specimens.
- RT-PCR and other molecular influenza assays are generally more expensive than other influenza tests.
- Some molecular influenza assays may not specifically identify all currently circulating influenza A virus subtypes. Depending on the test, a negative result for one influenza A virus subtype may not preclude infection with another influenza A virus subtype.
- Some molecular influenza assays being used are not FDA-cleared and an evaluation has not been performed to assess the accuracy of all available RT-PCR and molecular assays. A list of FDA-cleared tests is available in **Table FDA-cleared RT-PCR Assays and Other Molecular Assays for Influenza Viruses**.

- Rapid molecular influenza assays may have slightly lower sensitivity to detect influenza viruses compared to other RT-PCR and other molecular assays.

Table. FDA-cleared or FDA-authorized RT-PCR Assays and Other Molecular Assays for Influenza Viruses

| Products                                               | Manufacturer(s)             | Influenza Virus Type Detected | Influenza Virus Subtype(s) Differentiated | Other Respiratory Viruses Differentiated                                                                                                                                           | Acceptable Specimens <sup>[1]</sup>                                                                                                                                                                                           | Test Time <sup>[2]</sup> / Complexity <sup>[3]</sup> |
|--------------------------------------------------------|-----------------------------|-------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| ID NOW™ INFLUENZA A & B 2                              | Abbott                      | Influenza A and B             | None                                      | None                                                                                                                                                                               | Nasal Swab Direct or in VTM <sup>[4]</sup>                                                                                                                                                                                    | 0.25 h/ Waived                                       |
| Xpert Xpress Flu                                       | Cepheid                     | Influenza A and B             | A/2009 H1                                 | None                                                                                                                                                                               | Nasal or Nasopharyngeal swabs in VTM <sup>[4]</sup> or UTM <sup>[4]</sup>                                                                                                                                                     | 0.67 h/ Moderate                                     |
| Xpert® Xpress CoV-2/Flu/RSV plus (US-IVD Moderate)     | Cepheid                     | Influenza A and B             | None                                      | Respiratory Syncytial Virus and SARS-CoV-2                                                                                                                                         | Nasopharyngeal swab or anterior nasal swab in VTM <sup>[4]</sup> , UTM <sup>[4]</sup> or eNAT® <sup>[4]</sup>                                                                                                                 | <1.0 h/ Moderate                                     |
| cobas™ Influenza A/B + RSV                             | Roche Diagnostics           | Influenza A and B             | None                                      | RSV                                                                                                                                                                                | Nasopharyngeal swabs                                                                                                                                                                                                          | 0.33 h/ Waived                                       |
| Solana® Influenza A+B Assay                            | QuidelOrtho                 | Influenza A and B             | None                                      | None                                                                                                                                                                               | Nasal swabs and nasopharyngeal swabs in transport media* Nasal swabs and nasopharyngeal swabs in transport media* *BD/Copan VTM <sup>[4]</sup> , MicroTest™ M4, MicroTest™ M4RT, MicroTest™ M5, MicroTest™ M6, or Copan eSwab | 0.75 h/ Moderate                                     |
| Simplexa Flu A/B & RSV Gen II                          | Diasorin                    | Influenza A and B             | None                                      | RSV                                                                                                                                                                                | Nasopharyngeal swabs (in VTM <sup>[4]</sup> )                                                                                                                                                                                 | 1.5 h/ Moderate                                      |
| Simplexa™ COVID-19 & Flu A/B Direct                    | Diasorin                    | Influenza A and B             | None                                      | RSV and SARS-CoV-2                                                                                                                                                                 | Nasopharyngeal swabs (in VTM <sup>[4]</sup> )                                                                                                                                                                                 | <1.5 h/ Moderate                                     |
| VERIGENE® Respiratory Pathogens Flex Test              | Diasorin                    | Influenza A and B             | A/H1 (including H1 and 2009 H1), and A/H3 | Respiratory Syncytial Virus subtype A, Respiratory Syncytial Virus subtype B, Parainfluenza 1, 2, 3, and 4 virus, Human Metapneumovirus, Adenovirus, and Rhinovirus                | Nasopharyngeal swabs (in VTM <sup>[4]</sup> )                                                                                                                                                                                 | 2 h/ Moderate                                        |
| NxTAG® Respiratory Pathogen Panel                      | Diasorin                    | Influenza A and B             | A/H1, A/H3                                | Respiratory Syncytial Virus subtype A, Respiratory Syncytial Virus subtype B, Parainfluenza 1, 2, and 3 virus, Human Metapneumovirus, Rhinovirus, and Adenovirus                   | Nasopharyngeal swabs (in VTM <sup>[4]</sup> )                                                                                                                                                                                 | 3 h/ High                                            |
| Xpert Xpress Flu/RSV                                   | Cepheid                     | Influenza A and B             | None                                      | Respiratory Syncytial Virus                                                                                                                                                        | NPS, NS in VTM <sup>[4]</sup>                                                                                                                                                                                                 | 0.5 h/ CLIA Waived                                   |
| Cobas® Influenza A/B Assay                             | Roche Molecular Diagnostics | Influenza A and B             | None                                      | None                                                                                                                                                                               | NPS in VTM <sup>[4]</sup>                                                                                                                                                                                                     | <30 min/ CLIA Waived                                 |
| cobas® iat SARS-CoV-2, Influenza A/B & RSV Assay (EUA) | Roche Diagnostics           | Influenza A and B             | None                                      | SARS-CoV-2 and Respiratory Syncytial Virus                                                                                                                                         | Anterior nasal swab or NPS                                                                                                                                                                                                    | 0.33 h/ Waived                                       |
| Acucy® Influenza A&B                                   | Sekisui Diagnostics         | Influenza A and B             | None                                      | None                                                                                                                                                                               | NS direct or NPS                                                                                                                                                                                                              | 0.25 h/ Waived                                       |
| BIOFIRE® RESPIRATORY 2.1 (RP2.1) PANEL.                | BioMerieux                  | Influenza A and B             | H1, H1pdm09, H3                           | Adenovirus, Coronavirus (HKU1, NL63, 229E, OC43), Human Metapneumovirus, Human Rhinovirus/Enterovirus, Parainfluenza Virus (1, 2, 3, 4) Respiratory Syncytial Virus and SARS-CoV-2 | NPS in VTM <sup>[4]</sup>                                                                                                                                                                                                     | 0.75 h/ Moderate or High                             |
| BIOFIRE® Respiratory 2.1-EZ (RP2.1- EZ) Panel (EUA)*   | BioMerieux                  | Influenza A and B             | H1, H1pdm09, H3                           | Adenovirus, Coronavirus (HKU1, NL63, 229E, OC43), Human Metapneumovirus, Human Rhinovirus/Enterovirus, MERS-CoV, Parainfluenza Virus (1, 2, 3, 4) Respiratory Syncytial Virus      | NPS in VTM <sup>[4]</sup>                                                                                                                                                                                                     | 0.75 h/ Waived                                       |
| cobas® eplex Respiratory Pathogen Panel 2 (EUA)        | Roche Diagnostics           | Influenza A and B             | H1, H1pdm09, H3                           | Adenovirus, Coronavirus (229E, HKU1, NL63, OC43), Human Metapneumovirus, Human Rhinovirus/Enterovirus,                                                                             | NPS in VTM <sup>[4]</sup>                                                                                                                                                                                                     | 1.5 h/ Moderate or High                              |

| Products                                                                                                                    | Manufacturer(s)        | Influenza Virus Type Detected | Influenza Virus Subtype(s) Differentiated    | Other Respiratory Viruses Differentiated                                                                                                                                                          | Acceptable Specimens <a href="#">[1]</a>                                                              | Test Time <a href="#">[2]</a> / Complexity <a href="#">[3]</a> |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
|                                                                                                                             |                        |                               |                                              | Parainfluenza (1, 2, 3, 4), Respiratory Syncytial Virus (A, B) and SARS-CoV-2                                                                                                                     |                                                                                                       |                                                                |
| <b>QIAstat-Dx Respiratory SARS-CoV-2 Panel</b>                                                                              | Qiagen                 | Influenza A and B             | H1, H1pdm09, H3                              | Adenovirus, Coronavirus HKU1, NL63, 229E, and OC43 Human Metapneumovirus A+B, Human Rhinovirus/Enterovirus, Parainfluenza Virus 1, 2, 3, 4, Respiratory Syncytial Virus, SARS-CoV-2               | NPS in UTM <a href="#">[4]</a>                                                                        | 1 h/ Moderate                                                  |
| <b>BioCode Respiratory Pathogen Panel</b>                                                                                   | Applied BioCode Inc.   | Influenza A and B             | H1, H3, H1pdm09                              | Respiratory Syncytial Virus A/B, Parainfluenza viruses 1-4, Human Metapneumovirus A/B, Rhinovirus/Enterovirus, Adenovirus, Coronaviruses (229E, OC43, HKU1, NL63)                                 | NPS in VTM <a href="#">[4]</a> or UTM <a href="#">[4]</a>                                             | 5 h/ High                                                      |
| <b>FilmArray Pneumonia Panel</b>                                                                                            | (BioMérieux)           | Influenza A and B             | None                                         | Respiratory Syncytial Virus, Parainfluenza Virus, Human Metapneumovirus, Rhinovirus/Enterovirus, Adenovirus, Coronavirus                                                                          | Induced/ expectorated sputum, TRA, BAL or mini-BAL                                                    | 1 h/ High                                                      |
| <b>CDC Human Influenza Virus Real-Time RT-PCR Diagnostic Panel (Influenza A/B Typing Kit) <a href="#">[5]</a></b>           | CDC Influenza Division | Influenza A and B             | None                                         | None                                                                                                                                                                                              | NPS, NS, TS, NA, NW, NPS/TS, BAL, BW, TRA, sputum, lung tissue, viral culture                         | 4 h/ High                                                      |
| <b>CDC Human Influenza Virus Real-Time RT-PCR Diagnostic Panel (Influenza A Subtyping Kit) <a href="#">[5]</a></b>          | CDC Influenza Division | Influenza A                   | None                                         | None                                                                                                                                                                                              | NPS, NS, TS, NA, NW, NPS/TS, BAL, BW, TRA, sputum, lung tissue, viral culture                         | 4 h/ High                                                      |
| <b>CDC Human Influenza Virus Real-Time RT-PCR Diagnostic Panel (Influenza B Lineage Genotyping Kit) <a href="#">[5]</a></b> | CDC Influenza Division | Influenza B                   | Victoria-like lineage; Yamagata-like lineage | None                                                                                                                                                                                              | NPS, NS, TS, NA, NW, NPS/TS, viral culture                                                            | 4 h/ High                                                      |
| <b>CDC Human Influenza Virus Real-Time RT-PCR Diagnostic Panel (Influenza A/H5 Subtyping Kit) <a href="#">[5]</a></b>       | CDC Influenza Division | Influenza A                   | H5N1 (Asian lineage)                         | None                                                                                                                                                                                              | Respiratory specimens, viral culture, conjunctival swabs <a href="#">[6]</a>                          | 4 h/ High                                                      |
| <b>Panther Fusion® Flu A/B/RSV Assay</b>                                                                                    | Hologic                | Influenza A and B             | None                                         | Respiratory Syncytial Virus                                                                                                                                                                       | NPS UTM <a href="#">[4]</a> or VTM <a href="#">[4]</a>                                                | 2.5 h/ Moderate or High                                        |
| <b>Panther Fusion® SARS-CoV-2/Flu A/B/RSV Assay</b>                                                                         | Hologic                | Influenza A and B             | None                                         | SARS-CoV-2, Respiratory Syncytial Virus                                                                                                                                                           | Nasal swab or NPS UTM <a href="#">[4]</a> , eSTM, or VTM <a href="#">[4]</a>                          | 2.5 h/ Moderate or High                                        |
| <b>NxTAG® Respiratory Pathogen Panel</b>                                                                                    | Diasorin               | Influenza A and B             | H1/H1pdm09, H3                               | Adenovirus, Coronavirus (HKU1, NL63, 229E, OC43), Human Bocavirus, Human Metapneumovirus, Parainfluenza Virus (1, 2, 3, 4), Respiratory Syncytial Virus (A,B), Rhinovirus/Enterovirus             | NPS in VTM <a href="#">[4]</a>                                                                        | >3 h/ High                                                     |
| <b>NxTAG® Respiratory Pathogen Panel v2</b>                                                                                 | Diasorin               | Influenza A and B             | H1/H1pdm09, H3                               | Adenovirus, Coronavirus (HKU1, NL63, 229E, OC43), Human Bocavirus, Human Metapneumovirus, Parainfluenza Virus (1, 2, 3, 4), Respiratory Syncytial Virus (A,B), Rhinovirus/Enterovirus, SARS-CoV-2 | NPS in UTM <a href="#">[4]</a> , MicroTest™ M4RT™, or viral transport media (VTM) <a href="#">[4]</a> | 3 h/ Moderate or High                                          |
| <b>ALINITY m RESP-4-PLEX ASSAY</b>                                                                                          | Abbott                 | Influenza A and B             | None                                         | Respiratory Syncytial Virus and SARS-CoV-2                                                                                                                                                        | Nasopharyngeal swabs                                                                                  | 2 h/ Moderate or High                                          |
| <b>Lyra Influenza A + B Assay</b>                                                                                           | Quidel Corporation     | Influenza A and B             | None                                         | None                                                                                                                                                                                              | NS, NPS in MicroTest™ M4, MicroTest™M4-RT, MicroTest™M5, MicroTest™M6, or UTM <a href="#">[4]</a>     | 1.25 h/ High                                                   |

\*Note: this list may not include all FDA-cleared or FDA-authorized RT-PCR assays or other molecular assays for detection of influenza viruses in respiratory specimens. For a complete list of influenza diagnostic tests available through FDA 501(k) clearance or Emergency Use Authorization for use in clinical settings or for home use, see: <https://www.fda.gov/medical-devices/in-vitro-diagnostics/influenza-diagnostic-tests> [↗](#)

SOURCES AND PAGE INFO

**CONTENT SOURCE:**

[National Center for Immunization and Respiratory Diseases \(NCIRD\)](#)

**ABOUT THIS PAGE**

Published: October 1, 2024

Updated: April 30, 2026

This page was last updated on this date. Updates may include minor edits, image changes, or other modifications to page content.

Reviewed: April 30, 2026

The information on this page was last reviewed by subject matter experts to ensure accuracy.

**REFERENCES**

1. These specimen types are specified in product package inserts cleared by the U.S. Food and Drug Administration (FDA)
2. Test Time is inclusive of actual test time and is exclusive of transport, handling, laboratory run schedules and generating results. Timing may vary depending on extraction process used. Contact laboratory for expected turn-around time.
3. Clinical Laboratory Improvement Amendments require categorization of tests as waived, moderate or high complexity. Ref: [Categorization of Tests | CMS](#) 
4. VTM = viral transport media; UTM = universal transport media; eNAT® is FDA cleared Flu A RNA stabilization with implied use for specimens with suspected SARS-CoV-2.
5. Available only to qualified DoD laboratories and U.S. public health laboratories
6. Conjunctival swabs are currently approved under FDA enforcement discretion for testing with the CDC Influenza A/H5 subtyping assay.